ZCAS University

CCS1601 INTERNET OF THINGS FUNDAMENTALS

MID-SEMESTER TEST

1ST JUNE,2026

Time: 14:00 – 17:00 Hours

TIME ALLOWED: THREE HOURS

(plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

A university campus has 500 computers spread across 5 buildings. The network administrator notices frequent broadcast storms and wants to segment the network into smaller broadcast domains without replacing all cabling.

a) Explain the difference between a **switch** and a **router** in terms of broadcast domain handling. The administrator decides to use **VLANs**. Describe what a VLAN is and how it helps reduce broadcast traffic. If each building needs its own VLAN, how many **subnets** would be required if the university uses 172.16.0.0/16? Briefly justify. (5 marks)

A wearable fitness tracker monitors heart rate and step count. For normal data, the company stores it for long-term health trend analysis in the cloud. For **abnormal heart rate** (e.g., >140 bpm), an alert must reach the user's family member within 1 second.

b) Design a **hybrid cloud-fog architecture** that satisfies both requirements. Specify what processing happens at the wearable, fog node, and cloud. Between the wearable and fog node, would you choose **Bluetooth LE** or **Wi-Fi**? Justify your choice based on power consumption and range. Between the fog node and cloud, would you use **MQTT** or **HTTP**? Justify based on event-driven communication needs. (5 marks)

c)Write Python code that executes TEN numerical values and squares them. (8 marks)

d)Write a program of your choice using a WHILE LOOP. (6 marks)

e) Write the output of this code

```
species = ['dog', 'cat', 'shark', 'falcon', 'deer', 'tyrannosaurus rex']
i = 0
while i < 3:
    print(species[i])
    i = i + 1
```

(6 marks)

f) Describe the **OSI model** and list its seven layers in order from bottom to top. For each of the following devices, state the OSI layer at which it operates and explain its primary function:

- Router

- Switch
- Repeater

(10 marks)

(Total: 40 marks)

SECTION B: Attempt any THREE questions in this section.

Question 2

Scenario:
A startup wants to launch a mobile app for photo editing. They expect unpredictable user traffic, ranging from 100 to 10,000 users per hour. They have no IT staff to manage servers.

a) Which **cloud service model** (IaaS, PaaS, or SaaS) should they choose? Explain how **auto-scaling** (rapid elasticity) helps this startup handle traffic spikes. Name one **cloud provider** and one service they could use for this app. (10 marks)

b) A bank wants to move some operations to the cloud but must comply with strict financial regulations requiring customer data to stay within the country. They also need a dedicated environment for legacy applications.

i) Which **cloud deployment model** (public, private, or hybrid) best suits the bank? Explain why. (4 marks)

ii) Define **data sovereignty** and explain why it matters to the bank. (5 marks)

iii) Suggest one way the bank can encrypt data before sending it to the cloud. (1 mark)

(20 marks)

Question 3

A city deploys **connected traffic lights** that adjust timing based on real-time vehicle flow. Each intersection has a camera and a small computing device. The city wants to avoid sending video feeds to a central cloud due to network delays.

a) Draw a **three-layer architecture** (Cloud – Fog – Edge) for this system. Label each layer and describe the role of the fog layer. (10 marks)

b) Explain two specific **real-time decisions** a fog node at an intersection could make without contacting the cloud. (5 marks)

c) What is a potential **security risk** of processing traffic data at the fog layer? (1 mark)

(20 marks)

Question 4

Functions allow us to package functionality in a nice and readable way, reuse it without rewriting it, and make code modular and readable. A rule of thumb is that if you plan on using very similar code more than once, it may be worthwhile to write it as a reusable function.

(a) Write a Python program using the following functions showing indentation.

i. def (5 marks)

ii. if else and elif (5 marks)

(b) Write a Python program that defines a dictionary using the type (days of the week) (10 marks)

(20 marks)

Question 5

A smart warehouse uses 50 temperature sensors that send readings every 5 seconds to a local fog node (a Raspberry Pi or edge server). The fog node must:

1. Filter readings – discard any reading below -10°C or above +60°C (sensor faults).
2. Detect critical events – if temperature exceeds +40°C, send an immediate alert to the cloud (simulated).
3. Aggregate normal data – every 30 seconds, compute the average temperature of all valid readings and send only the average to the cloud (to save bandwidth).
4. Simulate network issues – randomly drop 5% of messages sent to the cloud to mimic real network packet loss.

You are to write a Python program that simulates this fog node.

(20 marks)

END OF EXAM